TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	85.1 ± 1.22
Correlation Stability	90.2 ± 1.66
Distribution Consistency	89.3 ± 1.90
Entropy Preservation	88.8 ± 0.75
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	83.2 ± 0.40
Privacy Risk	48.3 ± 1.00
Data Density	100.0 ± 0.01
Coverage Score	28.0 ± 3.22
Feature Accuracy	92.1 ± 1.51
Novelty Score	0.5 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.187	-2.559	5.7%	OK
TB	3.299	4.047	+0.748	22.7%	Warn
DB	1.486	1.833	+0.347	23.4%	Warn
Alkphos	290.576	260.644	-29.933	10.3%	Warn
Sgpt	80.714	62.341	-18.372	22.8%	Warn
Sgot	109.911	80.903	-29.007	26.4%	Drift
TP	6.483	6.512	+0.029	0.5%	OK
ALB	3.142	3.295	+0.153	4.9%	OK
A/G Ratio	0.947	0.949	+0.002	0.2%	OK
Selector	1.286	1.283	-0.003	0.2%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	85.1 ± 1.22
Correlation Stability (*2)	90.2 ± 1.66
Distribution Consistency (*3)	89.3 ± 1.90
Entropy Preservation (*4)	88.8 ± 0.75
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	83.2 ± 0.40
Privacy Risk (*7)	48.3 ± 1.00
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	28.0 ± 3.22

Feature Accuracy (*10)	92.1 ± 1.51
Novelty Score (*11)	0.5 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6931623931623931	0.6811965811965812	-0.005128205128205125	
F1	0.7971299319768493	0.8007301761332384	0.7960853617212179	-0.0010445702556314608	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6606837606837608	0.5991452991452992	0.6547008547008546	-0.005982905982905973	
F1	0.7639692470458219	0.7029462761017419	0.7574590917501476	-0.006510155295674369	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6803418803418804	0.6213675213675214	0.7008547008547008	0.020512820512820506	
F1	0.7876084221789155	0.72150669976497	0.8023519081480573	0.0147434859691418	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7051282051282051	0.7068376068376068	0	
F1	0.8279091624529537	0.8266734606474791	0.8279091624529537	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5957264957264957	0.7	-0.003418803418803418	
F1	0.8065231151809626	0.7048926028546825	0.8042906144504484	-0.0022325007305144416	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.6478632478632478	0.7119658119658119	0.00854700854700856	
F1	0.8108734553482932	0.7610936799050079	0.8199026539483839	0.00902919860009063 7	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6256410256410256	0.7025641025641025	0	
F1	0.7997903571549948	0.7362556041700637	0.8000902015911928	0.00029984443619804 9	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7111111111111111	0.6444444444444445	0.7094017094017094	-0.00170940170940174 78	
F1	0.8136666341240224	0.755758491508465	0.815518671279386	0.00185203715536378 39	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6615384615384616	0.6957264957264957	0.02136752136752136 8	
F1	0.7821343746683586	0.772932609943756	0.8057143222086811	0.02357994754032241	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5803418803418803	0.5461538461538462	-0.01282051282051280 3	
F1	0.5587432119380917	0.6382757985617826	0.5423025936682961	-0.01644061826979556 5	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	24	Male	1.0	0.2	189	52	31	8.0	4.8	1.5
2	45	Male	0.6	0.2	245	22	24	7.1	3.4	0.9
3	32	Male	32.6	14.1	219	95	235	5.8	3.1	1.1
4	66	Female	2.9	1.3	168	21	38	5.5	1.8	0.4
5	34	Male	6.2	3.0	240	1680	850	7.2	4.0	1.2
6	27	Male	0.7	0.2	243	21	23	5.3	2.3	0.7
7	29	Male	1.2	0.3	210	45	32	6.5	3.2	1.0
8	52	Female	0.5	0.1	150	30	25	7.0	4.0	1.1
9	38	Male	15.0	7.5	300	80	100	6.0	3.5	0.9
10	45	Female	2.0	0.8	190	50	60	5.8	2.9	1.2
11	60	Male	0.9	0.4	220	55	70	7.1	3.8	1.3
12	34	Female	12.5	6.0	180	40	50	6.3	3.1	1.0
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Male	2.2	1.0	610	17	28	7.3	2.6	0.55
2	35	Male	0.8	0.2	198	36	32	7.0	4.0	1.3
3	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
4	21	Male	18.5	9.5	380	390	500	8.2	4.1	1.0
5	66	Male	11.3	5.6	1110	1250	4929	7.0	2.4	0.5
6	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
7	29	Female	1.2	0.5	250	45	30	6.5	3.0	1.1
8	54	Male	0.9	0.3	220	50	40	7.2	3.4	1.2
9	37	Female	2.5	1.1	300	60	55	7.8	3.8	1.0
10	45	Male	0.8	0.2	180	35	25	5.8	2.9	0.9
11	62	Female	1.6	0.7	400	70	65	8.0	4.1	1.3
12	50	Male	3.2	1.5	350	55	50	6.7	3.2	1.0
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	30	Female	0.8	0.2	198	30	58	5.2	2.8	1.1
2	42	Male	1.0	0.3	154	38	21	6.8	3.9	1.3
3	55	Male	4.4	2.9	230	14	25	7.1	2.1	0.4
4	32	Male	23.0	11.3	300	482	275	7.1	3.5	0.9
5	32	Male	15.9	7.0	280	1350	1600	5.6	2.8	1.0
6	58	Male	0.9	0.2	1100	25	36	7.1	3.5	0.9
7	29	Female	1.2	0.4	210	25	30	6.0	3.0	1.0
8	37	Male	0.9	0.3	180	20	25	5.5	2.8	0.9
9	45	Female	2.5	1.0	300	40	50	7.0	3.5	1.1
10	52	Male	1.8	0.6	250	30	35	6.5	3.2	1.0
11	60	Female	0.7	0.2	220	15	20	5.8	2.5	0.8
12	34	Male	3.1	1.5	400	50	60	7.5	4.0	1.2
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	0.8	0.2	286	21	27	7.1	4.0	1.2
2	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
3	48	Female	0.9	0.2	173	26	27	6.2	3.1	1.0
4	21	Male	1.0	0.3	142	27	21	6.4	3.5	1.2
5	32	Male	22.7	10.2	290	322	113	6.6	2.8	0.7
6	66	Male	11.3	5.6	1110	1250	4929	7.0	2.4	0.5
7	29	Female	1.5	0.6	210	35	45	6.0	3.2	0.53
8	54	Male	4.2	1.8	320	65	75	7.2	3.9	1.1
9	38	Female	0.9	0.3	180	40	50	6.5	3.1	0.8
10	45	Male	2.5	1.1	250	30	60	5.8	2.5	0.67
11	62	Female	3.1	1.4	400	70	80	7.0	3.5	0.9
12	27	Male	1.2	0.5	190	25	35	6.2	2.8	1.0
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Male	23.0	11.3	300	482	275	7.1	3.5	0.9
2	48	Female	0.9	0.2	173	26	27	6.2	3.1	1.0
3	45	Male	2.3	1.3	282	132	368	7.3	4.0	1.2
4	26	Male	1.0	0.3	163	48	71	7.1	3.7	1.0
5	60	Male	8.9	4.0	950	33	32	6.8	3.1	0.8
6	40	Male	1.1	0.3	230	1630	960	4.9	2.8	1.3
7	29	Female	1.2	0.5	210	45	50	6.5	3.2	1.1
8	37	Male	2.5	1.1	300	60	70	7.0	3.5	1.3
9	54	Female	0.8	0.3	180	30	40	5.8	2.9	0.9
10	45	Male	3.1	1.4	250	80	90	6.0	3.0	1.2
11	62	Female	1.5	0.6	220	55	65	6.7	3.3	1.0
12	39	Male	0.9	0.2	190	25	30	5.5	2.8	0.8
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	75	Male	10.6	5.0	562	37	29	5.1	1.8	0.5
2	47	Male	0.9	0.2	192	38	24	7.3	4.3	1.4
3	55	Male	0.8	0.2	482	112	99	5.7	2.6	0.8
4	49	Female	0.8	0.2	158	19	15	6.6	3.6	1.2
5	34	Male	6.2	3.0	240	1680	850	7.2	4.0	1.2
6	32	Male	30.5	17.1	218	39	79	5.5	2.7	0.9
7	29	Female	1.5	0.5	210	40	30	6.5	3.2	1.1
8	52	Male	12.3	6.5	450	75	90	7.0	3.8	0.7
9	37	Female	0.9	0.3	150	20	25	5.8	2.5	0.8
10	61	Male	3.4	1.2	300	55	60	6.2	3.0	1.0
11	44	Female	18.1	9.0	600	80	100	7.5	4.0	0.9
12	25	Male	0.6	0.2	180	30	20	6.0	2.8	1.2
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	1.6	0.8	218	18	20	5.9	2.9	0.96
2	42	Female	0.5	0.1	162	155	108	8.1	4.0	0.9
3	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
4	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
5	32	Male	15.9	7.0	280	1350	1600	5.6	2.8	1.0
6	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
7	29	Female	0.9	0.3	150	25	30	6.5	3.2	0.8
8	54	Male	2.5	1.0	200	40	50	7.2	3.8	1.1
9	37	Female	1.2	0.5	180	30	40	6.8	3.4	0.9
10	45	Male	3.0	1.5	220	35	45	7.5	3.6	1.0
11	62	Female	0.7	0.2	160	28	32	6.0	2.9	0.7
12	39	Male	1.5	0.6	210	50	55	7.0	3.5	0.8
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Male	0.7	0.2	165	55	87	7.5	4.6	1.58
2	40	Male	0.6	0.1	171	20	17	5.4	2.5	0.8
3	66	Male	17.3	8.5	388	173	367	7.8	2.6	0.5
4	32	Male	15.9	7.0	280	1350	1600	5.6	2.8	1.0
5	43	Male	0.8	0.2	192	29	20	6.0	2.9	0.9
6	70	Male	1.3	0.4	358	19	14	6.1	2.8	0.8
7	25	Female	1.5	0.5	150	30	40	6.5	3.5	1.85
8	34	Male	0.9	0.3	200	45	50	7.0	4.0	1.75
9	50	Female	2.1	0.8	300	60	70	6.2	3.0	1.2
10	29	Male	0.7	0.2	180	25	35	5.8	2.8	1.0
11	42	Female	1.8	0.6	250	55	65	7.1	4.2	1.5
12	38	Male	0.6	0.1	220	40	55	6.8	3.1	1.1
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	47	Male	0.9	0.2	192	38	24	7.3	4.3	1.4
2	35	Male	1.6	0.7	157	15	44	5.2	2.5	0.9
3	54	Female	22.6	11.4	558	30	37	7.8	3.4	0.8
4	32	Male	18.0	8.2	298	1250	1050	5.4	2.6	0.9
5	28	Male	0.6	0.2	159	15	16	7.0	3.5	1.0
6	48	Male	5.0	2.6	555	284	190	6.5	3.3	1.0
7	29	Female	1.2	0.5	210	45	50	6.0	3.2	0.7
8	37	Male	3.5	1.2	300	50	60	6.5	3.0	0.9
9	45	Female	2.1	0.9	250	40	55	5.8	2.8	0.8
10	52	Male	4.0	1.5	400	80	70	7.0	3.5	0.9
11	60	Female	1.5	0.6	180	30	40	5.5	2.5	0.7
12	34	Male	2.8	1.1	220	55	65	6.2	3.1	0.8
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Male	0.8	0.2	158	29	39	6.0	2.2	0.5
2	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
3	57	Male	1.4	0.7	470	62	88	5.6	2.5	0.8
4	46	Male	20.0	10.0	254	140	540	5.4	3.0	1.2
5	33	Male	3.4	1.6	186	779	844	7.3	3.2	0.7
6	66	Male	11.3	5.6	1110	1250	4929	7.0	2.4	0.5
7	25	Female	1.0	0.4	150	22	30	6.5	3.0	1.2
8	41	Male	2.5	1.0	250	35	45	7.0	3.5	1.0
9	34	Female	0.8	0.2	180	28	32	6.8	3.2	1.1
10	50	Male	3.0	1.5	300	40	50	7.2	3.8	0.9
11	29	Female	1.2	0.5	200	25	35	6.7	3.1	1.3
12	38	Male	2.0	0.9	220	30	40	7.5	3.6	1.0
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3